
Assessing Simple Versus Complex Restoration Strategies for Industrially Disturbed Forests

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Abstract

Assessing the community-level consequences of ecological restoration treatments is essential to guide future restoration efforts. We compared the vegetation composition and species richness of restored sites that received a range of restoration treatments and those of unrestored sites that experienced varying levels of disturbance. Our study was conducted in the industrially degraded landscape surrounding Sudbury, Ontario, Canada. The Great Lakes–St. Lawrence Forest once present in this area was degraded through logging, mining, and smelting activities beginning in the late 1800s until restoration of the most visibly degraded areas began in 1974. Restoration treatments ranged from simple abiotic enhancements to complex, multistage revegetation treatments using native and non-native species, which included fertilizing, spreading of ground dolomitic limestone, understory seeding, and tree planting. Canonical correspondence analysis was used to determine which restoration treatments explained

differences in the community structure among sites. We found that native understory vascular species richness was similar in restored sites that received more complex restoration treatments and unrestored sites that were mildly disturbed; however, the role of planted trees and non-native species in the restored communities remains unclear. Understory vascular seeding played a key role in determining community composition of vascular understory and overstory communities, but the time since restoration commenced was a more important factor for nonvascular communities because they received no direct biotic enhancements. The use of non-native species in the vascular seed mix seems to be slowly encouraging the colonization of native species, but non-natives continue to dominate restored sites 25 years after restoration began.

Key words: complexity, disturbance, monitoring, multivariate, ordination, pollution, restoration, reference system.

Introduction

The restoration and conservation of forest biota has become increasingly important in the face of large-scale forest degradation (Urbanska et al. 2000; Young 2000). Ecologists confronted with the task of reconstructing a native forest must take into account several considerations including the cost and timespan for change and, consequently, the intensity and nature of restoration strategy. Valuable knowledge can be gleaned from past efforts at ecological restoration. In some instances, removal of the degrading influence may be all that is needed to encourage recovery (Berger 1993; Brown 1999), particularly in the presence of a viable seed source and a long-term time frame. However, if the system has crossed a threshold at which point the disturbance affects critical ecosystem processes, then a more active restoration approach may be required to promote successional processes operating on a time scale that is compatible with human needs (Gilbert & Anderson 1998; Golley 1999).

Existing methods of active restoration may be broadly categorized according to the level of management intervention. One-dimensional restoration efforts (e.g., involving single abiotic variables, single species, or single applications) are common but may be limited in their efficacy because recovery processes are multidimensional (e.g., Van de Koppel et al. 1997; Nystrom et al. 2000; Adams et al. 2002). Observed nonlinear or threshold responses to management and environmental factors could indicate that ecosystems exist in alternate stable states that are resilient to simplistic, one-dimensional restoration strategies (Hobbs & Norton 1996; Suding et al. 2004). Therefore, more complex, multidimensional restoration efforts that employ a combination of methods (e.g., site preparation, establishment of plantation forests, and control of non-native species) may provide a more rapid and effective intervention strategy (Fulé et al. 2002). Similarly, one-dimensional assessment of the recovery of ecosystems (e.g., that exclusively target recovery of one species) does not serve to reflect state of recovery of the community as a whole (Anand et al. 2002; Anand & Desrochers 2004). Even the commonly studied community parameters, such as biomass, abundance, and richness (Holl & Cairns 2002), may provide an incomplete, or even misleading, picture of community structure when studied alone (Lawton et al. 1998; Anand & Desrochers 2004). Increasingly, multivariate methods are being applied to

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restoration assessments to better understand recovery dynamics and to identify the relationship between community structure and environmental variables (de Gruchy et al. 2001; Wiegand & Felinks 2001; Moynahan et al. 2002; McLachlan & Bazely 2003).

A more complex forest restoration technique emerging from tropical research involves not only removing abiotic constraints but also establishing plantation forests (Jussi et al. 1995; Haggard et al. 1997; Carnevale & Montagnini 2002; Feyera et al. 2002). The establishment of tree plantations in degraded areas may facilitate regeneration of native species that could not otherwise establish by removal of abiotic constraints alone (Lugo 1992a). Both native and non-native plantations can act as catalyzers for the invasion of native species by changing understory microclimatic conditions, increasing vegetation structural complexity, and developing litter and humus layers (Parrotta et al. 1997; Cusack & Montagnini 2004). However, the use of non-native species in ecological restoration is contentious (d'Antonio & Meyerson 2002; Hoddle 2004; Ewel & Putz 2004) in large part due to the potential for these non-native species to become invasive and hence problematic in surrounding areas. Perhaps, the best known example is that of *Robinia pseudoacacia*, which is considered a pest in several areas where it was once used to improve microclimatic conditions on mined lands (Ashby 1987).

The efficacy of removing abiotic constraints versus establishing tree plantations to serve as "nurse crops" in temperate regions has not been studied as extensively. However, it has been shown that small fields in close proximity (160 m) to forest edge can quickly become diverse forests without supplemental tree planting (Kruse & Groninger 2003). Mixed plantations of native and non-native tree species established at low densities (120 trees/ha) further facilitated the invasion of light-seeded species within 50 years in small old-field bottomlands in Kentucky (Shear et al. 1996). Non-native plantations of Austrian pine in France were shown to act as a nurse stand, enabling the return of native broad-leaved trees and herbaceous species (Vallauri et al. 2002). This study confirmed the importance of low-density plantations to improve their role as "seed traps" and advocated tree thinning to promote the true restoration of native biodiversity (Vallauri et al. 2002). Bossuyt and Hermy (2000) suggest that, in addition to plantation, forest understory species should also be introduced, and Cox et al. (2004) recommend sowing native, understory seed mix followed by irrigation and rolling of seeds into soil. Hartman and McCarthy (2004) further recommend the eradication of a dominant, non-native understory species to increase success of native tree seedlings.

The objective of our study is to assess restoration strategies of differing complexity and compare these to unrestored sites that have been subjected to varying levels of disturbance. We examine species richness and composition of nonvascular, vascular, and overstory vegetation.

We hypothesize that increasing restoration complexity will produce forests more similar in species richness (but not necessarily composition) to a mildly disturbed, unrestored forest and substantially different from severely disturbed, unrestored forest. We also hypothesize that the complexity of restoration strategies will be reflected in all forest strata. This work is important for defining the most appropriate strategies for restoration and for guiding intervention practices in restored ecosystems. It also contributes to the ongoing debate of whether natural systems should be left to recover on their own or whether there should be human intervention of any kind (Brown 1999; Kirmer & Mahn 2001; Prach et al. 2001).

Methods

Vegetation surveys were conducted in naturally recovering and restored sites within the industrially damaged landscape of Sudbury, Ontario, Canada (lat 46.5°N, long 81.0°W). Sudbury was originally surrounded by Great Lakes–St. Lawrence Forest, a transitional forest zone between the Boreal coniferous forest to the north and the Carolinian deciduous forest to the south, characterized by Red pine (*Pinus resinosa* Aiton) and White pine (*Pinus strobus* Linnaeus) (Courtin 1994; Winterhalder 1996, 2000). The landscape is characterized by a rolling topography of hills with shallow soils on Canadian Shield bedrock. Temperatures range from 30°C in summer to –40°C in winter. It receives an average rainfall of 71.3 mm in the summer, and the ground is covered in snow from December to March. The Sudbury landscape has been degraded by logging, mining, and smelting activities since the late 1800s. In 1972, increased government air pollution controls resulted in the decommissioning of one of Sudbury's smelting complexes, and shortly thereafter, restoration of the most visibly degraded areas began (Potvin & Negusanti 1995). Different restoration treatments were applied to 10,000 ha of degraded land in and around Sudbury ranging from (1) fertilizing with mineral 6-24-24 N-P-K fertilizer applied at 390 kg/ha (Lautenbach 1985); (2) manual and aerial spreading of ground dolomitic limestone; (3) seeding of non-native grass and legume species; and (4) planting of 1- to 2-year-old primarily native tree seedlings (*Quercus rubra*, *Picea glauca*, *Pinus banksiana*, *P. resinosa*, *P. strobus*) (Winterhalder 1995, 1996). Non-native species in the understory seed mixture were graminoids (*Poa compressa* 15%, *Poa pratensis* 15%, *Phleum pratense* 20%, *Agrostis gigantea* 20%, and *Festuca rubra* 10%) and leguminous forbs (*Lotus corniculatus* 10% and *Trifolium hybridum* 10%) (Lautenbach et al. 1995).

Seven restored study sites (sites 1A–7A) were sampled within a 4-km radius just east of Sudbury (Fig. 1). These sites were all severely affected by the smelting activities. They varied with respect to the complexity of the restoration techniques (Table 1). A reclamation map provided by the City of Sudbury was used to determine which mixture of

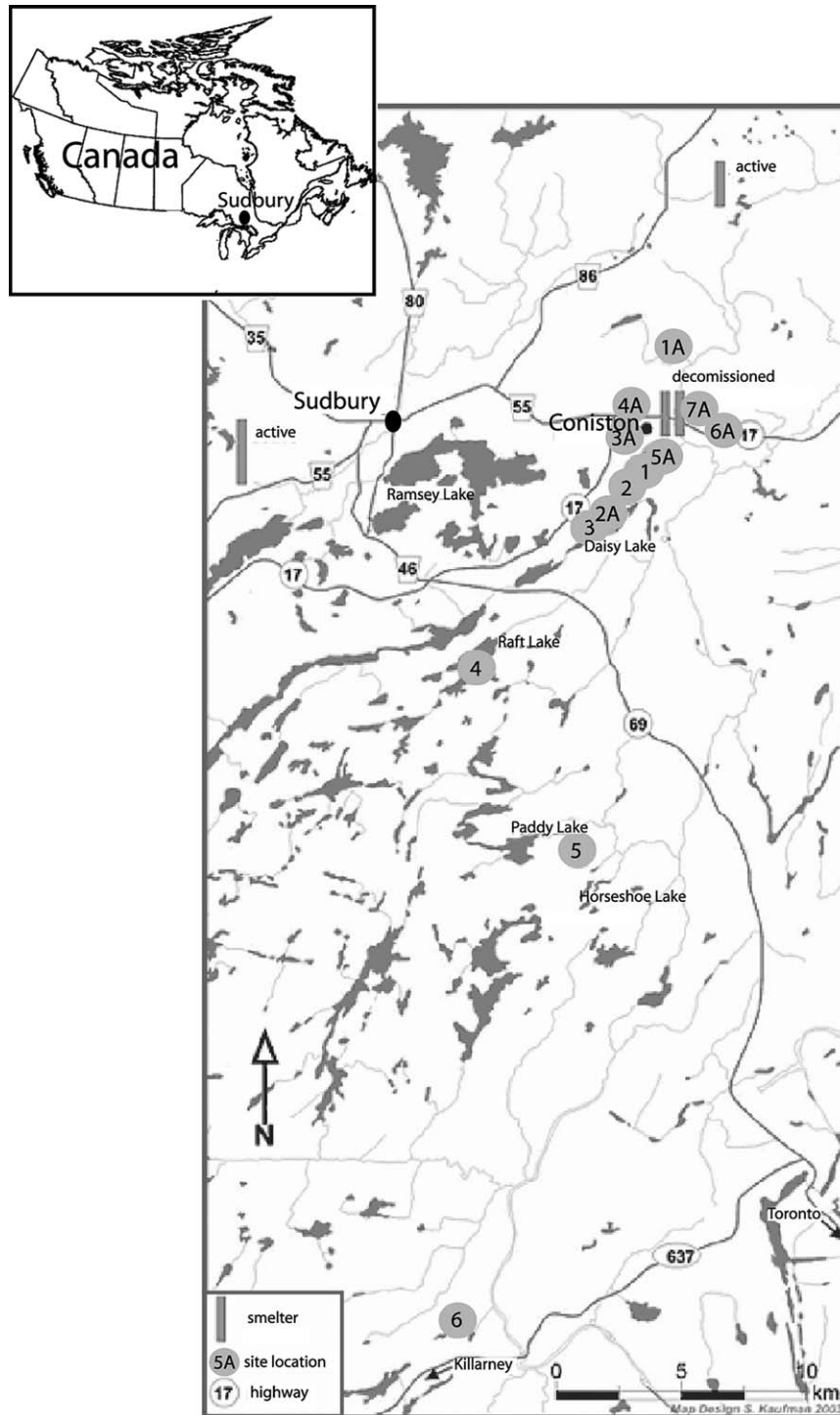


Figure 1. Map showing the locations of the six unrestored sites (sites 1–6) and the seven restored sites (sites 1A–7A) relative to the decommissioned smelting complex near Sudbury, Ontario, Canada.

treatments could be assumed to have affected our sampling areas. Some of the resulting restoration techniques were in fact unintended. For example, some “limed-only” areas were left as such due to the fact that the City of Sudbury’s reclamation program ran out of seed (K. Winterhalder 2005,

Laurentian University, personal communication). However, these “accidents” proved for an interesting scientific study to compare various restoration techniques. Six naturally recovering (unrestored) sites were used for comparison. These unrestored sites are part of the Canadian Ecological

Table 1. Distance from decommissioned smelter (km), type and year(s) of treatment, time span of treatment, tree species planted, density of planted trees, and species richness for unrestored sites (sites 1–6) and restored sites (sites 1A–7A).

Site	Distance to Smelter (km)	pH	Restoration Techniques and Year(s) of Implementation	Timespan of Treatment (Years)	Tree Species Planted	Density of Trees Planted (Number of Trees/Area)	Total Species Richness
1A	3.5	6.28	Limed (1997)	1	None	NA	19
2A	3.6	6.18	Aerial limed + seeded (1994)	5	<i>Pinus banksiana</i>	0.0300	21
3A	3.3	5.97	Tree planted (1995)				
			Fertilized (1979)	18	<i>P. banksiana</i>	0.0162	20
			Tree planted (1983, 1996)		<i>Pinus resinosa</i>	0.0366	
					<i>Pinus strobus</i>	0.0057	
4A	3.7	6.65	Limed + fertilized + seeded (1979)	19	<i>P. banksiana</i>	0.0032	35
			Tree planted (1983, 1984, 1997)		<i>P. resinosa</i>	0.0031	
					<i>P. strobus</i>	0.0142	
5A	0.7	6.36	Limed (1983) + fertilized + seeded (1984)	17	<i>P. banksiana</i>	0.0265	26
			Tree planted (1986, 1995, 1997, 1998, 1999)		<i>P. resinosa</i>	0.0375	
					<i>P. strobus</i>	0.029	
					<i>Quercus rubra</i>	0.0085	
6A	1.9	6.31	Limed (1979, 1981, 1988)	18	<i>Picea glauca</i>	0.0777	26
			Fertilized + seeded (1981, 1988)		<i>P. banksiana</i>	0.0960	
			Tree planted (1983, 1989, 1990, 1994, 1996, 1997)		<i>P. resinosa</i>	0.0692	
					<i>P. strobus</i>	0.119	
7A	1.7	6.37	Limed (1979, 1990)	19	<i>Robinia pseudodacia</i>	0.0014	39
			Fertilized + seeded (1990)		<i>P. glauca</i>	0.0348	
			Tree planted (1987, 1988, 1992, 1993, 1997)		<i>P. banksiana</i>	0.1016	
					<i>P. resinosa</i>	0.0403	
					<i>P. strobus</i>	0.0037	
					<i>Q. rubra</i>	0.0037	
1	1.7	4.27	None	NA	None	NA	8
2	2.0	4.27	None	NA	None	NA	8
3	4.2	4.56	None	NA	None	NA	10
4	11.9	4.53	None	NA	None	NA	20
5	16.1	4.04	None	NA	None	NA	22
6	35.8	4.19	None	NA	None	NA	43

Monitoring and Assessment Network (<http://www.eman-rese.ca>) and lie along the anthropogenic disturbance gradient, starting near the decommissioned Coniston smelter and proceeding in a southerly direction for approximately 36 km (sites 1, 2, & 3 fall within a roughly 4-km radius of the smelter and are considered “severely disturbed,” sites 4 & 5 “moderately disturbed,” and site 6 “mildly disturbed”; Fig. 1).

In all sites, understory vegetation was sampled during the summer of 2003 along 50- to 100-m transects perpendicular to south-facing slopes to minimize differences due to light availability. Transect length varied among sites according to site area; however, cumulative species area–species richness curves appeared to reach their asymptote in all sites. Understory vascular and nonvascular vegetation was sampled using the Braun Blanquet method of estimating cover abundance in 1×1 -m contiguous quad-

rats along each transect. Understory species were identified in the field according to Chambers et al. (1996), and voucher specimens of vascular and nonvascular species were identified to species in the lab with the aid of Britton and Brown (1970), Newmaster et al. (1998), Conrad and Redfearn (1979), and Brodo et al. (2001). Overstory species were defined as woody perennials greater than 1.5 m tall, due to the stunted nature of overstory in this degraded landscape, and were sampled for tree density in 5×5 -m contiguous quadrats. The vegetation was partitioned into three strata for analyses: vascular understory, nonvascular understory, and overstory.

Richness and percent cover of vegetation strata were compared between all restored and unrestored sites using log-likelihood tests. Restored sites were classified according to restoration complexity such that sites that received a subset of all available restoration treatments (sites 1A,

2A, & 3A) were “simple,” and sites that received all available restoration treatments (sites 4A, 5A, 6A, & 7A) were “complex.” The Wilcoxon signed rank nonparametric test was used to assess the effect of restoration complexity on species richness and Pearson’s product-moment correlation coefficient was calculated to relate restoration time span to species richness at the site level.

Species composition at restored and unrestored sites was related to environmental variables and restoration treatments by performing canonical correspondence analyses (CCA). CCA is an ordination method that identifies differences in species composition among sites and relates species abundances to measured variables by constraining site and species placements in ordination space to be linear combinations of the measured variables (Legendre & Legendre 1998). We used CANOCO (ter Braak 1998) to perform CCA on the species abundances of each stratum using the following seven variables: distance to smelter, number of years of separate limings, fertilized or not fertilized, seeded or not seeded, overall timespan of restoration activities, pH, and density of planted trees. The effects of tree plantations on overstory species composition were further investigated by performing pair-wise correlations between the densities of the major planted trees and sampled trees using JMP Statistical Discovery Software (version 5, release 5.1).

Results

Richness

There were significant differences between the restored and unrestored sites for native and non-native classes of vegetation in terms of species richness ($G = 58.57$, $p < 0.0001$) and percent cover ($G = 108.37$, $p < 0.0001$; Fig. 2).

However, the highest overall richness values were observed in the mildly disturbed, unrestored site (site 6 had 43 species), and a site that received a complex restoration treatment (site 7A had 39 species; Fig. 3). Vascular plant richness was in fact higher in site 7A than in site 6, due only in part to the direct addition of non-native vascular species in the restoration treatment. To further examine differences in restored sites, we examined effects of restoration complexity and restoration timespan on species richness of native and non-native classes of vegetation. Restoration time span was not correlated with species richness in any of the vegetation classes (native nonvascular $r = 0.261$, $p = 0.572$; native vascular $r = 0.537$, $p = 0.213$; non-native vascular $r = 0.425$, $p = 0.342$; total richness $r = 0.637$, $p = 0.124$). However, restoration complexity, where restored sites were classified as having received either simple (sites 1A, 2A, & 3A) or complex (sites 4A, 5A, 6A, & 7A) restoration, resulted in significant differences in total species richness ($Z = -1.96$, $p = 0.049$) and near-significant differences in non-native vascular species richness ($W = 6.0$, $p = 0.057$). Restoration complexity did not produce differences in native nonvascular ($Z = -1.80$, $p = 0.072$) species richness or native vascular ($Z = 0.36$, $p = 0.718$) species richness. A normal approximation of the Wilcoxon test was used, and the Z statistic was reported when richness data had tied values (because exact p values cannot be computed if there are tied values).

Many weedy, non-native vascular species (that were not directly introduced in the seed mixture) colonized the restored sites that received complex restoration: *Achillea millefolium*, *Cirsium arvense*, *Dianthus armeria*, *Hieracium* spp., *Medicago sativa*, *Melilotus alba*, *Oenothera biennis*, *Rumex acetosella*, *Sonchus arvensis*, *Taraxacum officinale*, *Verbascum thapsus*, and *Vicia cracca* (classified

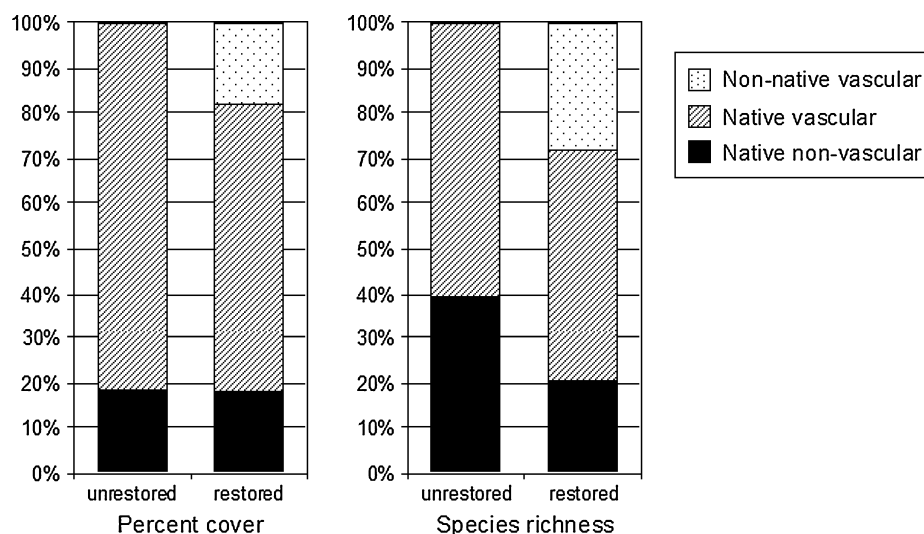


Figure 2. Percent cover and species richness of unrestored sites (sites 1–6) and restored sites (sites 1A–7A). Species are categorized as native vascular, non-native vascular, and native nonvascular.

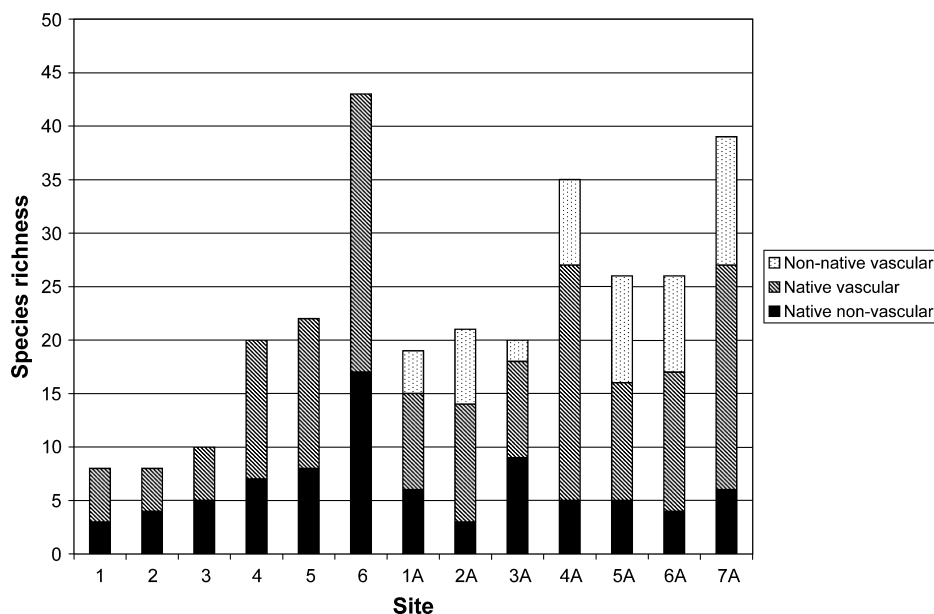


Figure 3. Species richness of nonvascular and vascular understory species for the unrestored sites (sites 1–6) and restored sites (sites 1A–7A).

in SWSS 1998). However, these complex restored sites were also colonized by native forest understory species that were absent from the unrestored sites, such as *Anaphalis margaritacea*, *Aster umbellatus*, *Calamagrostis canadensis*, *Deschampsia flexuosa*, *Erigeron canadensis*, and *Ledum groenlandicum*. These are largely light-tolerant species, which likely thrived in the relatively open canopy that was characteristic of the restored sites, in part, due to the stunted overstory. Several shade-tolerant, native forest species were in turn only present in the unrestored sites, such as *Corylus cornuta*, *Gaultheria procumbens*, *Oryzopsis asperifolia*, and *Diphysastrum digitatum*, due in part to the fuller canopy.

Sites that received the most simplistic restoration techniques (sites 1A, 2A, & 3A) had similar numbers of vascular species as the moderately disturbed sites (sites 4 & 5) but had fewer nonvascular species (Fig. 3). Site 3A, the only site that was fertilized but did not receive the non-native vascular plant seed mixture had the highest nonvascular species richness and the lowest vascular species richness of the restored sites, and only had two non-native species present. The species richness of the nonvascular communities of the mildly disturbed, unrestored sites was much higher than that of any of the restored sites and included bryophyte species *Brachythecium populeum*, *Brachythecium rutabulum*, *Pleurozium schreberi*.

Canonical Correspondence Analyses

Measured environmental variables controlling the vascular plant composition are fairly well represented by the CCA axes (Fig. 4a), as indicated by the consistent placement of sites and species in ordination space in both the

CCA and a correspondence analysis (CA; not shown). Axis 1 (27.8%) differentiates between restored sites and naturally recovering sites that are least disturbed and is highly correlated with the environmental variable “distance from smelters” (Table 2). Axis 2 (19.3%) differentiates between sites that received different restoration strategies and also differentiates unrestored sites with varying disturbance levels. Sites that received the most complex restoration strategies and sites that are least disturbed are associated with more vascular understory species; however, the assemblages are different, and they are maximally separated along axis 2. Along both the axes, sensitive, shade-tolerant species such as *Cornus canadensis*, *Aralia nudicalis*, *Trientalis borealis*, and *Maianthemum canadense* are associated with the least disturbed site (site 6), and species present in the seed mixture are associated with sites receiving the most complex restoration (sites 5A, 6A, & 7A). Tree saplings in the understory that are primarily associated with site 6 are *C. cornuta*, *Acer rubrum*, and *Picea glauca*, whereas planted species such as *Pinus banksiana* and *Pinus resinosa* are associated with the restored sites 5A, 6A, and 7A and with site 6 along axis 2. *Deschampsia flexuosa* and *Betula papyrifera*, a native grass and tree species that are commonly found together in the Sudbury region (Sinclair 1996), are grouped in ordination space together with restored sites receiving simple restoration (sites 1A & 3A). *Gaultheria procumbens* and *Pteridium aquilinum* appear to be somewhat isolated from other species and sites in the ordination diagram; they are only found at the least disturbed sites 4, 5, and 6. *Ledum groenlandicum* is strongly associated with highly disturbed, unrestored sites 1 and 2 along both axes because its relative contribution to these species-poor sites was large; however, it was also found at

Table 2. Eigenvalues, % cumulative variance contribution, and weighted variable correlations for first three canonical axes of the CCA.

	Vascular Plants			Nonvascular Plants			Overstory		
	Axis 1	Axis 2	Axis 3	Axis 1	Axis 2	Axis 3	Axis 1	Axis 2	Axis 3
Eigenvalues	0.780	0.542	0.313	0.480	0.247	0.122	0.798	0.721	0.355
Cumulative % variance									
Of species	27.8	47.1	58.2	41.2	62.5	72.9	35.0	66.6	82.2
Of species-environment	35.8	60.6	75.0	51.7	78.3	91.3	40.6	77.3	95.3
Variables									
Distance	0.984	-0.112	-0.017	0.961	-0.180	-0.046	0.981	0.100	-0.013
No. of years of limings	-0.626	-0.548	-0.441	-0.189	0.330	0.716	-0.390	0.453	0.619
Fertilized	-0.658	-0.431	-0.083	-0.350	-0.471	0.401	-0.590	0.245	0.095
Seeded	-0.648	-0.639	-0.027	-0.148	0.124	0.843	-0.454	0.609	0.185
Time span	-0.700	-0.404	-0.187	-0.419	-0.477	0.471	-0.581	0.150	0.142
Density	-0.590	-0.525	-0.395	-0.208	0.046	0.542	-0.498	0.463	0.611
pH	-0.785	-0.371	-0.085	-0.497	-0.147	0.557	-0.673	0.225	0.010

restored sites 6A and 7A and is associated with these sites along axis 1.

The relationships between nonvascular plants and the measured environmental variables differed from those of the other strata. The results of the CCA for the nonvascular plants showed that the first axis explained a large amount (41.2%) of variation among sites but that only “distance from smelter” was highly correlated with this axis (Table 2). The nonvascular communities show site differentiation along this first axis, whereby sites that received the most complex restoration technique are closest to the least disturbed sites (Fig. 4b). However, along axis 2, some of the restored sites that received complex techniques (sites 3A & 4A) were closest to site 6 and associated with *Polytrichum juniperinum* and *Cladonia chlorophaea*, whereas site 5A was farthest from site 6 and was associated with *Pohlia nutans* and *Cladonia bacillaris*. Site 6 was strongly associated with *Stereocaulon tiomentosum* and *B. rutabulum* along both axes. Highly disturbed, unrestored sites (sites 1 & 2) and restored sites with both simplistic and complex restoration (sites 1A, 2A, & 7A) were clustered in ordination space but were not strongly associated with any nonvascular species in particular along either axis.

CCA on the overstory data did not match the ordering of sites and species produced by CA alone, suggesting that some unmeasured variables may play an important structuring role. Restored and unrestored sites are not as well differentiated, but unrestored sites that were relatively less disturbed (sites 4, 5, & 6) are clearly distinguishable from other sites along axis 2 (Fig. 4c). The seeding variable showed the highest correlation with this second axis (Table 2). Site 6 is isolated in ordination space largely due to the species exclusively present at this site such as *C. cornuta*. However, the overstory communities of all restored sites were not consistently distinguishable from those of unrestored sites with low disturbance due to the dominant success of the native pines (*P. banksiana* and *P. resinosa*) used for overstory restoration. This is also

reflected in the high correlations between the densities of the major planted and sampled trees: *P. banksiana* ($r^2 = 0.96, p = 0.0008$) and *P. resinosa* ($r^2 = 0.74, p = 0.05$). The pair-wise correlations were not significant for *Pinus strobus* ($r^2 = 0.02, p = 0.97$), *Quercus rubra* ($r^2 = -0.255, p = 0.58$), and *P. glauca* ($r^2 = -0.195, p = 0.68$).

Discussion

We found that increased complexity of the restoration strategy at a particular site was primarily reflected as an increase in non-native species richness and did not restore native vascular species, nonvascular species, or overstory richness to low disturbance reference levels. Consistent with our first hypothesis, the most complex restoration strategies have similar species richness to the mildly disturbed reference site; however, they have not served to recreate a similar species assemblage. Although there have been previous suggestions that forests exposed to small-scale disturbances that are located relatively close to remnant forest patches can reestablish native species richness within 40 years (McLachlan & Bazely 2001, 2003), studies of large-scale forest disturbances predict that the native plants will take many decades (Bratton et al. 1994) or centuries (Duffy & Meier 1992; McIntyre et al. 1994; Meier et al. 1995) to reach historic levels. It appears that although species richness has been restored to mildly disturbed reference levels relatively quickly in Sudbury's damaged landscape, rebuilding species composition will take more time or require further restoration effort. Winterhalder (1996) predicted that “areas treated in different years may end up with very different patterns and densities of woody plant cover, despite overall conditions.” Although we found no significant effects of the overall time span of restoration, we suggest that more work involving both survey and historical data, but also predictive ecological modeling, may be useful to examine this question in more detail.

Ordinations showed that restored sites can be distinguished from unrestored sites by quantifying the composition of the vascular or nonvascular understory communities but not as clearly for the overstory community. Differences in restored and unrestored overstories may be minimized at the present time due to the plantation of primarily native species and reduced growth rate of overstory species. The recovery of understory vascular communities was not consistent among all sites, and ordination showed a complexity gradient among restored sites, with those sites receiving complex techniques separated from those receiving simple techniques. Sites receiving complex treatments had understory vascular species compositions that were more similar to those of mildly disturbed sites than restored sites that received more simplistic treatments. The reverse was true for overstory and nonvascular species, which falsifies our second hypothesis that the complexity of restoration strategies would be reflected in all forest strata. Effects of the different restoration treatments do not follow a clear trend in the nonvascular communities of our study, which is consistent with the lack of temporal trend in disturbed bryophyte communities' compositions found by Ross-Davis and Frego (2002). Anand et al. (2005) showed that vascular plant richness was significantly correlated with other strata for the naturally recovering sites but not for the restored sites, likely due to the fact that the restoration techniques only targeted vascular communities.

Liming on its own has been determined to have a "trigger effect" in the severely damaged landscape of Sudbury (see Winterhalder 1996 and references therein) leading to rapid colonization of several species, which otherwise could not establish due to the highly acidic conditions of the soil. We found that *Populus tremuloides* was not present in the one site that we assumed had not been limed. This supports the idea that some species could be used as an indicator species for various treatments. Winterhalder (1996) also found that *P. tremuloides* was absent in nonlimed areas but managed to colonize limed areas very well. He suggests that because of the "trigger factor" of liming, other treatments may not be absolutely essential. For example, he suggests that the act of seeding may not be necessary, at least in terms of recolonization of woody species such as *P. tremuloides* and *Salix* spp.

We found, however, that the act of seeding the understory proved to be an important restoration technique for distinguishing between sites in the CCA analyses. The impact of a viable seed source on the structuring of understory communities has proven to be very important in other studies that considered "distance to continuous forest" (McLachlan & Bazely 2003) and "functional landscape connectivity" (Lindenmayer et al. 2002). Understory seeding should increase the rate of establishment of diversity, and we indeed found that the site that was not seeded was dominated by a well-established population of *Vaccinium angustifolium* (site 3A). Also, the site that received the most complex restoration treatment (site 7A)

had 13 native vascular species present in its understory that were not found in the understory of the mildly disturbed reference site (site 6) and seven non-native species that were not introduced in the seed mixture. This suggests that the non-native seeded species may be serving as a nurse-crop to aid the establishment of other native and non-native species.

Tree planting showed species-specific success in terms of "return on investment." Planting of *Pinus resinosa* and *Pinus banksiana* appeared successful in that planting densities were correlated with actual densities. However, plantings of *Pinus strobus*, *Quercus rubra*, and *Picea glauca* were not found to be as successful. Although these species were planted as part of the restoration, they were planted in smaller numbers and in more selected sites than *P. resinosa* and *P. banksiana*. For example, at site 6A, the density of *P. strobus* was 5-fold less than that of either *P. resinosa* or *P. banksiana* alone. Despite differences in planting densities, this also suggests that a reexamination of species-specific criteria for restoration using tree planting may be required. For example, *P. strobus* may not have been a suitable restoration species, given that it is highly sensitive to SO₂ pollution (Freedman & Hutchinson 1980; Amiro & Courtin 1981) and was only found in sites 5 and 6, which are at 16 and 36 km, respectively, from the decommissioned smelters. Nevertheless, as noted in other short-term assessments of restoration models (McLachlan & Bazely 2001, 2003; de Souza & Batista 2004), it is probable that differences in forest compositional structure attributable to restoration strategies will take a long time to appear, given the long time scale of ecological processes; therefore, long-term monitoring is also necessary.

Although diversity measures, such as overall vascular richness, are frequently used as indicators of forest recovery, they have been criticized for being too coarse and ignoring important changes in underlying species composition (Lawton et al. 1998). Our results indicate that even a more refined measure of diversity, *native* understory vascular species richness, is inadequate in characterizing the effect of the restoration strategy. McLachlan and Bazely (2001) also found that diversity of native vascular species did not differ between restored and undisturbed reference sites in deciduous forests in southern Canada. Similarly, Lugo (1992b) found that species richness in the understory of a mahogany plantation was similar to that of a paired secondary forest with a large number of native tree species present. However, Lugo (1992b) found important differences in the functioning of plantation and secondary growth forests in terms of root density, nutrient retranslocation, and leaf litter. Measuring ecosystem function directly using biomass or other fitness measures that may vary with level of disturbance (e.g., height of tree species; Tucker & Anand 2003) may improve our comparison of the restoration strategies because apparently successful communities in terms of composition may not be sustainable in the long term (Ewel 1987).

McLachlan and Bazely (2001) have used specific understory herbaceous species as indicators of regeneration, and others have shown the ability of understory herbs, individually and grouped as functional groups, to indicate disturbance associated with land use (Moffatt & McLachlan 2004). We found that site 6 was isolated in ordination space largely due to the species exclusively present at this site such as *Corylus cornuta*, which is characteristic of moist soils rich in calcium and nitrogen (Ringius & Sims 1997), suggesting the potential of indicator species to also serve as a rudimentary monitoring tool in this system. However, not all understory species can serve as indicators of recovery based on their relative abundance alone (Dufrene & Legendre 1997; Anand & Desrochers 2004). Sensitive, shade-tolerant species such as *Dryopteris marginalis* and *Waldsteinia fragarioides*, present only at the relatively undisturbed site 6 (0.18 and 0.26%), and *Maianthemum racemosum*, present at site 6 and the site that received the most complex restoration treatment (0.63 and 0.06%), may be too rare for reliable detection during sampling and may have difficulty becoming established even if habitat conditions are suitable (Gigon & Langenauer 1998). Other woodland species that are present at the least disturbed site, such as *C. cornuta*, *Aralia nudicaulis*, and *Trientalis borealis*, and constitute a considerable amount of the understory (15.74, 6.79, and 3.33%, respectively) are also absent from all other sites. *Maianthemum canadense* and *Gaultheria procumbens*, which are present at two (4.5 and 13.10%) and three (0.1, 1.56, and 1.40%) of the naturally recovering sites, respectively, may be better to use as indicators.

Given that the vascular and nonvascular communities have responded to the restoration techniques so differently in the degraded landscape of Sudbury, we suggest that nonvascular species may also be useful as indicators of recovery, and research should be focused in this area. The presence of lichens at the sites close to the pollution source is an indicator of improving air quality (Beckett 1995). *Polytrichum* spp. are often used for monitoring heavy metal deposition (Tyler 1990), and *Polytrichum juniperinum* was found at sites 1 and 2 along with *Pohlia nutans*, which is considered to be highly tolerant to copper and nickel (Freedman & Hutchinson 1980; Tyler 1990; Brodo et al. 2001). *Cladonia chlorophaea* is the only lichen present at site 1. This species has broad requirements and can grow on thin soil over rock outcrops (Thomson 1967). Various lichen species, such as *Cladonia deformis* and *Stereocaulon* spp., can grow on *P. nutans* once this species is established (Amiro & Courtin 1981); it is possible that *C. chlorophaea* appeared in site 1 after the cover of *P. nutans* was established. The mildly disturbed site (site 6) had the highest nonvascular plant diversity of all sites (restored and unrestored) and was strongly associated with *Stereocaulon tomentosum* and *Brachythecium rutabulum*, which suggests that these species may be indicators for this site.

Complex restoration strategies appear to increase the rate of recovery; however, this study suggests that restora-

tions involving plantations using non-native species will encourage the establishment of native species, but the non-native vascular species will persist and dominate understory communities for at least a quarter century after restoration commences. Restoration efforts via planting should not only target vascular communities but also nonvascular communities to increase bryophyte and lichen richness and to conserve relationships between taxonomic groups (Anand et al. 2005). Steel et al. (2004) showed that the structure of bryophyte communities did not differ from that of vascular communities in undisturbed sites; however, our study showed that even the species abundances of nonvascular communities were not being restored with complex restoration strategies, whereas they were for the vascular communities. Because increasing the complexity of restoration strategies can have inevitably unpredictable effects on richness and composition of native versus non-native species in plant communities, models of recovery and restoration should take into account multiple and alternative target states, particularly in light of the fact that we are still at a stage where the role of native versus non-native species in restoration is unknown (Ewel & Putz 2004; Tucker & Anand 2004).

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